

———— // ZPWR-MIDI-FX // ————

ZPWR MIDI-FX - Block Catalog

v0.3.0

MODULE REFERENCE

block catalog

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MenkeTechnologies

———— MENKETECHNOLOGIES ————

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Catalog at a glance

111 blocks shipped by **zpwr-midi-fx** across 10 categories, including 0 component-level analog-circuit models and 45 control / modulation (CV-badge) sources. This is the per-plugin subset of the shared zpwr-patch-core catalog.

Registry MIDI 111
Output jack Audio 0 MIDI 111 CV 0

By category (block count, largest first):

Sequencing.....	39	Control.....	5
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Reading this catalog (badge key)

An icon glyph precedes each block name; then its badges, then **in**: its input jacks and **out**: its output-jack type, then the description:

- > **[CAT]** category - **AMP** Amp, **ANA** Analog, **CTL** Control, **DLY** Delay, **DST** Distortion, **DYN** Dynamics, **ENV** Envelope, **FLT** Filter, **FX** Effect, **HRM** Harmony, **MOD** Modulation, **OSC** Oscillator, **PIT** Pitch, **PRB** Probability, **REV** Reverb, **RTE** Routing, **SEQ** Sequencing, **SPC** Spectral, **STR** Stereo, **TUN** Tuning, **UTL** Utility, **VOI** Voicing.
- > **[CV]** a control / modulation source (outputs CV, not audio).
- > **[CIRCUIT]** a component-level analog-circuit model (per-sample nodal / Newton solve).
- > **[FX]** **[SYNTH]** **[MIDI]** which **plugin(s)** ship the block - **zpwr-fx**, **zpwr-synth** (gets the whole audio pack) and/or **zpwr-midi-fx**.
- > **in**: the block's input jacks, and **out**: its output-jack signal type - **out: Audio**, **out: MIDI** (note stream) or **out: CV** (control / modulation).
- > **icon**: the glyph before each block name shows its kind at a glance:

	Oscillator		Filter		Delay
	Reverb		LFO / mod		Envelope
	Dynamics		Stereo		Pitch
	Spectral		Harmony		Sequencer
	Routing		Voicing		Effect
	Control		Probability		Tuning
	Utility		Guitar amp		Pedal / stompbox
	Speaker cabinet		Analog circuit		Other

MIDI (zpw-midi-fx) (111 blocks)

Control (5)

⌚ **ATProc** [CTL][CV][MIDI] in: Audio out: MIDI

Scales aftertouch, optionally converting it to a CC.

⌚ **BendProc** [CTL][CV][MIDI] in: Audio out: MIDI

Scales and/or inverts incoming pitch-bend.

⌚ **CC2Note** [CTL][CV][MIDI] in: Audio out: MIDI

A watched CC crossing a threshold triggers a note.

⌚ **CCMap** [CTL][CV][MIDI] in: Audio out: MIDI

Remaps a CC number and scales/offsets its value.

⌚ **Note2CC** [CTL][CV][MIDI] in: Audio out: MIDI

Emits a CC from each note (pitch or velocity).

Dynamics (10)

⌚ **Accent** [DYN][MIDI] in: Audio out: MIDI

Boosts velocity on every Nth note.

⌚ **Humanize** [DYN][MIDI] in: Audio out: MIDI

Adds random timing and velocity jitter for a human feel.

⌚ **Ramp** [DYN][MIDI] in: Audio out: MIDI

Velocity ramp over a run of notes, ascending or descending.

⌚ **VelClip** [DYN][MIDI] in: Audio out: MIDI

Clamps velocity between Min and Max.

⌚ **VelCompress** [DYN][MIDI] in: Audio out: MIDI

Velocity compressor: velocities above Threshold are pulled toward it by Ratio (1 = off, 8 = hard), narrowing the dynamic range smoothly instead of hard-clamping like VelClip.

T **VelCurve** [DYN][MIDI] in: Audio out: MIDI

Reshapes velocity through a power curve.

T **VelInvert** [DYN][MIDI] in: Audio out: MIDI

Flips the velocity scale around its midpoint (soft <-> loud) by Amount, so your accents become ghost notes and the ghosts become accents - a velocity mirror, blendable to taste.

T **VelKey** [DYN][MIDI] in: Audio out: MIDI

Keyboard velocity tracking: scales each note's velocity by how far its pitch sits from a Center note, Tilt setting how strongly and which way - higher notes louder (or softer), the dynamic key-tracking of real instruments.

T **Vellength** [DYN][MIDI] in: Audio out: MIDI

Note duration scales with velocity: hard notes ring for Max ms, soft notes for Min (set Max < Min to make accents staccato) - the played articulation where dynamics shape note length.

T **Velocity** [DYN][MIDI] in: Audio out: MIDI

Scales and offsets note velocity, with a modulation amount.

Expression (3)

☐ **Fall** [FX][MIDI] in: Audio out: MIDI

Jazz fall-off: on release the pitch sweeps down by Depth over Time while the note rings, then the note releases and the bend snaps back to centre. The note-off is delayed to the end of the fall.

☐ **MidiVibrato** [FX][MIDI] in: Audio out: MIDI

A pitch-bend LFO that fades in after an Onset delay while any note is held - the delayed, played-in vibrato of a string or voice rather than an instant wobble. Rate in Hz, Depth as bend amount.

☐ **Scoop** [FX][MIDI] in: Audio out: MIDI

Bends up into each struck note from Depth below, resolving to pitch over Time - the vocal/brass scoop, a pitch-bend ramp that ends at centre so no bend is left hanging.

Harmony (21)

⌵ **AutoInvert** [HRM] [MIDI] in: Audio out: MIDI

Cycles the inversion of each struck chord: every trigger rotates which of the lowest notes jump an octave (by Rotate), so a repeated chord keeps climbing through its inversions - movement ChordVoicing's static shapes don't give.

⌵ **Bassline** [HRM] [MIDI] in: Audio out: MIDI

Follows the chord with a mono bass voice: the lowest held note dropped Octaves down, retriggered whenever the lowest note changes. Passes the original notes through and adds the bass beneath them.

⌵ **Chord** [HRM] [MIDI] in: Audio out: MIDI

Turns each incoming note into a chord (165 types) with inversion, spread, octave doubling and strum.

⌵ **ChordLock** [HRM] [MIDI] in: Audio out: MIDI

Quantizes every note to the nearest tone of a fixed chord (Root + Type), tighter than Scale's scale-quantize - locks an improvisation to a chord's notes.

⌵ **ChordMem** [HRM] [MIDI] in: Audio out: MIDI

Each key plays a stored chord shape (semitone intervals from node config).

⌵ **ChordProg** [HRM] [MIDI] in: Audio out: MIDI

Each note triggers the next chord in a stored root progression (node config).

⌵ **ChordVoicing** [HRM] [MIDI] in: Audio out: MIDI

Re-voices each note into a jazz shape (open / shell / quartal / spread).

⌵ **Counterpoint** [HRM] [MIDI] in: Audio out: MIDI

Adds a second voice in contrary motion - when the melody rises the counter voice falls and vice-versa - quantised to Root/Scale. Interval sets the starting separation.

⌵ **Diatonic** [HRM] [MIDI] in: Audio out: MIDI

Transposes by scale DEGREES within Root/Scale (a third, a fifth...) instead of by raw semitones like Transpose - the shift always lands back in key. Mod Amt adds a modulatable degree offset.

⌵ **DiatonicChord** [HRM] [MIDI] in: Audio out: MIDI

Auto-harmoniser: builds the in-key chord on every note by stacking diatonic thirds (triad, 7th...) along Root/Scale, so single-finger melodies become correct chords in the key.

⌵ **Drone** [HRM] [MIDI] in: Audio out: MIDI

Doubles every note with a fixed pedal root.

⌵ **Glissando** [HRM] [MIDI] in: Audio out: MIDI

Fills a stepwise run between consecutive notes.

⌵ **Harmonize** [HRM] [MIDI] in: Audio out: MIDI

Adds up to three fixed-interval harmony voices to each note.

⌵ **Invert** [HRM] [MIDI] in: Audio out: MIDI

Mirrors pitch around a pivot note.

⌵ **MidiCluster** [HRM] [MIDI] in: Audio out: MIDI

Adds symmetric tone-cluster voices above and below each note.

⌵ **MidiFold** [HRM] [MIDI] in: Audio out: MIDI

Folds out-of-range notes back inside a Low..High window.

⌵ **NeoRiemann** [HRM] [MIDI] in: Audio out: MIDI

Neo-Riemannian triad transforms: builds a major/minor triad on each note and applies P (parallel), L (leading-tone) or R (relative) - the maximally-smooth chord moves of late-Romantic and film harmony, each sharing two common tones with the input.

⌵ **Octave** [HRM] [MIDI] in: Audio out: MIDI

Doubles each note up and/or down by whole octaves.

⌵ **Scale** [HRM] [MIDI] in: Audio out: MIDI

Quantizes incoming notes to the nearest degree of the chosen root and scale.

⌵ **Transpose** [HRM] [MIDI] in: Audio out: MIDI

Shifts every note by a fixed semitone amount plus a modulatable offset.

⌵ **VoiceLead** [HRM] [MIDI] in: Audio out: MIDI

Smooth voice leading: places each note's pitch class in the octave nearest the previous output, so a line or chord progression moves by the smallest interval. Range caps the leap; past it the original octave is kept.

Modulation (13)

🎵 **CCGlide** [MOD] [MIDI] in: Audio out: MIDI

Portamento for a CC: smooths the watched CC toward each newly received value over Time, emitting interpolated CC each block - de-zippers a stepped controller, where MidiSlew only glides the internal scalar.

🎵 **CCLFO** [MOD] [MIDI] in: Audio out: MIDI

Continuous CC LFO generator (sine/triangle/saw/square) emitted on a chosen CC at a free Rate - automates a synth parameter over MIDI, where MidiLFO drives only the internal mod-matrix scalar.

🎵 **CCSeq** [MOD] [MIDI] in: Audio out: MIDI

Tempo-synced step CC sequencer: one CC value per synced step cycling an 8-step shape (overridable via the node's config list) - a modulation sequencer in the CC domain, where StepSeqMidi sequences notes.

🎵 **Generative** [MOD] [MIDI] in: Audio out: MIDI

Probabilistic melodic random walk synced to a rate.

🎵 **MidiEnv** [MOD] [MIDI] in: Audio out: MIDI

ADSR envelope triggered by the note gate, on the scalar control output.

🎵 **MidiLFO** [MOD] [MIDI] in: Audio out: MIDI

Low-frequency modulation source (sine/tri/saw/square) on the scalar control output.

🎵 **MidiRandom** [MOD] [MIDI] in: Audio out: MIDI

Generates random notes within a pitch range at a synced rate.

🎵 **MidiSampleHold** [MOD] [MIDI] in: Audio out: MIDI

Samples and holds the incoming control value on each note.

🎵 **MidiSlew** [MOD] [MIDI] in: Audio out: MIDI

Smooths (glides) the control value over a set time.

🎵 **PitchDrift** [MOD] [MIDI] in: Audio out: MIDI

Nudges each note's pitch by a small random detune up to +/-Cents (per-note, polyphonic) - the gentle oscillator instability of analog gear, applied to pitch where Humanize only scatters timing and velocity.

🎵 **PressSwell** [MOD] [MIDI] in: Audio out: MIDI

Generates a channel-pressure (aftertouch) swell that rises to Max over Rise while any note is held, snapping back to zero on release - an automatic crescendo for pads and strings, where ATProc only transforms incoming aftertouch.

~ **RandOctave** [MOD] [MIDI] in: Audio out: MIDI

Randomly shifts notes by up to +/- Range octaves with a chance.

~ **VelToCC** [MOD] [MIDI] in: Audio out: MIDI

Turns each note's velocity into a CC value emitted just before the note, so a synth's filter or level opens with how hard you played - dynamics routed to expression, where Note2CC sends pitch.

Probability (1)

⋮ **Chance** [PRB] [CV] [MIDI] in: Audio out: MIDI

Probabilistic gate: lets each note through with the set probability.

Routing (11)

◀ **Channel** [RTE] [MIDI] in: Audio out: MIDI

Reassigns notes to a fixed MIDI channel.

◀ **FixedNote** [RTE] [MIDI] in: Audio out: MIDI

Forces every note to a single fixed pitch (drum / trigger).

◀ **KeySwitch** [RTE] [MIDI] in: Audio out: MIDI

Splits the keyboard at a note; one zone passes, the other is filtered.

◀ **Merge** [RTE] [MIDI] in: Audio out: MIDI

Passes both note-stream inputs through, merged into one stream.

◀ **MidiLatch** [RTE] [MIDI] in: Audio out: MIDI

Holds notes after release until the next note arrives.

◀ **NoteFilter** [RTE] [MIDI] in: Audio out: MIDI

Passes only notes inside a key and velocity window.

◀ **NoteLimit** [RTE] [MIDI] in: Audio out: MIDI

Polyphony cap: never lets more than Max notes sound at once, stealing the oldest (FIFO note-off) when a new note would exceed the limit - tames dense chords and runaway generators.

—< **PolyChan** [RTE] [MIDI] in: Audio out: MIDI

Round-robins each new note to the next of Voices MIDI channels from Base, spreading a polyphonic part across a multitimbral rig - one note per channel, the classic voice-allocation split (distinct from MPEConvert's per-note MPE channels).

—< **Split** [RTE] [MIDI] in: Audio out: MIDI

Keyboard split: notes below the split point transpose by Low, the rest by High.

—< **Thin** [RTE] [MIDI] in: Audio out: MIDI

Note-density limiter: drops any note-on arriving within MinGap ms of the last one passed, decluttering a busy stream or fast repeats - distinct from NoteLimit, which caps simultaneous polyphony. A dropped note's off is dropped too.

—< **Transformer** [RTE] [MIDI] in: Audio out: MIDI

Rule: notes within a range are transposed and velocity-scaled.

Sequencing (39)

■ ■ ■ **Appoggiatura** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Leaning grace note: each struck note first sounds a long auxiliary (Interval away) that takes Time from the principal, which then sustains - the expressive long grace, unlike the quick Flam or Mordent.

■ ■ ■ **Arp** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Arpeggiator: cycles held notes through a pattern at a synced rate with gate, octaves, swing and step count. Latch holds the chord so it keeps arpeggiating after the keys are released (the next key starts a new chord).

■ ■ ■ **Cascade** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Each struck note fans out into a timed run of Steps notes climbing (or falling) the scale from it - an arpeggiated flourish fired per key, unlike Arp which cycles the whole held chord.

■ ■ ■ **ChordArp** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Cthulhu-style chord arpeggiator: each key builds a full chord (Type) and the result is arpeggiated upward across Octaves at a synced Rate - one finger plays a moving arpeggio of the whole chord. Releasing the key stops its run.

■ ■ ■ **Drunk** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Random-walk melodic sequencer: each synced Rate step nudges the pitch by a random +/-Step semitones, bounded within +/-Range of the lowest held note - Brownian melody anchored to the chord, unlike MidiRandom's memoryless picks.

■ ■ ■ **Echo** [SEQ] [CV] [MIDI] in: Audio out: MIDI

MIDI delay: repeats notes at a synced time with velocity feedback decay.

■ ■ ■ **EuclidMel** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Euclidean melody: spreads Pulses evenly across Steps (Bjorklund) and, on each hit, advances a cursor through the held chord - the rhythm is Euclidean and the pitch walks the chord, unlike SeqEuclid which stabs all held notes per pulse.

■ ■ ■ **Flam** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Precedes every note with a quiet grace note, delaying the main hit so the grace leads it - the classic drum flam.

■ ■ ■ **GateArp** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Cthulhu's arp gate lane: walks the held chord upward across Octaves while the note length cycles a per-step staccato/legato pattern, so the rhythm breathes - short stabs then sustained notes. Gate scales the pattern (>1 overlaps into the next step).

■ ■ ■ **MidiBrianBrain** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Brian's Brain MIDI sequencer: an 8x8 three-state grid (ready / firing / dying) evolves under Brian's Brain rules - a ready cell ignites only with exactly two firing neighbours, firing cells pass to dying, dying cells reset. Each clock step plays a note for every firing cell in the next column (row maps to a scale degree from Root/Scale); a full 8-step sweep advances one generation. The refractory state keeps it restless and sparkly where Game of Life settles. Density sets the random fill. The MIDI-note counterpart to the BrianBrain CV block.

■ ■ ■ **MidiGameOfLife** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Conway's Game of Life MIDI sequencer: an 8x8 cell grid evolves under the B3/S23 rule; each clock step reads the next column and plays a note for every live cell (row maps to a scale degree from Root/Scale), and a full 8-step sweep advances one generation. Gliders, blinkers and still-lives become ever-shifting melodic and harmonic patterns. Density sets the random fill when the grid (re)seeds. The MIDI-note counterpart to the GameOfLife CV block.

■ ■ ■ **MidiLangton** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Langton's Ant MIDI melody: a single turmite walks an 8x8 grid - turning right on a white cell, left on a black one, flipping each cell as it leaves, then stepping forward. Each clock advances the ant Steps cells and plays one note for its current row (mapped to a scale degree from Root/Scale), so the melody scribbles chaotically then locks into the famous periodic 'highway'. Emergent order from one trivial rule - a single moving agent, not a population.

■ ■ ■ **MidiMarkov** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Learns the note-to-note transition statistics of what you play (a 12x12 pitch-class chain) and, on each synced step, walks the chain to improvise a new line in the same style. Memory fades old transitions.

■ ■ ■ **MidiQuantize** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Snaps note starts to a synced grid by a strength amount.

■ ■ ■ **MidiTranceGate** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Chops the held notes with a 16-step on/off Pattern at a synced Rate: every 'on' step retriggers all held notes, gated to a fraction of the step. The rhythmic gate behind the trance sound - pattern-driven, where Roll is a fixed pulse.

■ ■ ■ **MidiTuring** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Turing-machine sequencer (Music Thing style): a circular shift register clocks once per step and the held notes fire when the read bit is 1. Change is the probability the recycled bit flips - 0 locks an evolving loop, 1 fully randomises, between = slowly mutating melodies.

■ ■ ■ **Mordent** [SEQ] [CV] [MIDI] in: Audio out: MIDI

One-shot ornament: each struck note plays principal -> auxiliary -> principal at the attack, then sustains, the single-flick counterpart to the continuous Trill.

■ ■ ■ **NoteLength** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Forces a fixed note duration (gate length).

■ ■ ■ **NoteOffDelay** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Extends each note's gate by delaying its note-off.

■ ■ ■ **OctaveArp** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Cthulhu's arp octave lane: arpeggiates the held chord while a per-step octave-offset Pattern leaps the line up and down by whole octaves (alternating, staircase, bounce...) instead of climbing monotonically like StepArp.

■ ■ ■ **PatternGate** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Gates note-ons through a 16-step on/off pattern (bitmask).

■ ■ ■ **Polymeter** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Two step lanes of different lengths (LenA, LenB) clock together over the held notes, phasing against each other so the combined pattern only repeats after lcm(LenA, LenB) steps - instant polymetric movement from one chord.

■ ■ ■ **RatchetArp** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Cthulhu's arp repeat lane: walks the held chord upward across Octaves while a per-step pattern retriggers selected steps multiple times (capped by Max), packing rolls into the line. Unlike SeqRatchet's fixed count, the repeat count varies per step.

■ ■ ■ **Ricochet** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Accelerating bouncing retriggers with decaying velocity.

■□□ **Roll**[SEQ][CV][MIDI] in: Audio out: MIDI

Drum roll / buzz: retriggers every held note at Rate for as long as it is held, each hit gated to a fraction of the interval.

■□□ **SeqEuclid**[SEQ][CV][MIDI] in: Audio out: MIDI

Euclidean rhythm gate: spreads Pulses evenly across Steps (rotatable) at a synced rate.

■□□ **SeqRatchet**[SEQ][CV][MIDI] in: Audio out: MIDI

Retriggers each note Count times within a synced step.

■□□ **Shift**[SEQ][CV][MIDI] in: Audio out: MIDI

Delays every note by Delay ms (groove nudge / lay-back), pushing note-ons and offs back by the same amount so durations are preserved - the per-track timing offset of a groove sequencer.

■□□ **SkipArp**[SEQ][CV][MIDI] in: Audio out: MIDI

Cthulhu's arp skip lane: arpeggiates the held chord upward across Octaves, but each step has a Skip probability of being dropped to a rest, thinning the pattern differently every cycle for generative movement.

■□□ **Spiral**[SEQ][CV][MIDI] in: Audio out: MIDI

Transposing MIDI delay: each repeat is delayed by Time and shifted by Interval semitones with velocity decaying by Feedback, so one note spins off a climbing or falling staircase of echoes - unlike Echo, which repeats the same pitch.

■□□ **StepArp**[SEQ][CV][MIDI] in: Audio out: MIDI

Cthulhu-style step arpeggiator: walks the held chord strictly upward across Octaves octaves at a synced Rate, accenting the first step of each cycle - the rolling, octave-jumping arp of Xfer Cthulhu's sequencer. Distinct from Arp's mode-cycling: this climbs the chord in order.

■□□ **StepSeqMidi**[SEQ][CV][MIDI] in: Audio out: MIDI

Melodic step sequencer - semitone offsets (node config) clock the held root.

■□□ **Strum**[SEQ][CV][MIDI] in: Audio out: MIDI

Spreads a chord's notes over time, up or down, like a guitar strum.

■□□ **Swing**[SEQ][CV][MIDI] in: Audio out: MIDI

Pushes off-beat eighth-notes later for a swing/shuffle feel.

■□□ **TransArp**[SEQ][CV][MIDI] in: Audio out: MIDI

Cthulhu's arp transpose lane: arpeggiates the held chord upward across Octaves while a per-step semitone Pattern (fifth pedals, triad arps, scalar runs) is added on top, so the line outlines a melodic contour, not just the chord tones.

■ ■ ■ **Trill** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Keyboard trill: while a note is held, alternates rapidly between it and the note Interval semitones above, at Rate.

■ ■ ■ **Tuplet** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Snap starts to a Triplets-per-beat grid (3 = triplets, 5 = quintuplets) by Strength, for a triplet or odd-tuplet feel where MidiQuantize only snaps to the straight grid.

■ ■ ■ **Turn** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Gruppetto turn: a four-step ornament at the attack - upper auxiliary, principal, lower auxiliary, then the principal held. Time sets each step, Interval the neighbour distance.

■ ■ ■ **VelArp** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Cthulhu's arp velocity lane as a ramp: arpeggiates the held chord while velocity sweeps from From to To across Steps then resets, so every pass swells or fades without a per-step editor.

Tuning (3)

└ **JustTune** [TUN] [MIDI] in: Audio out: MIDI

Adaptive just intonation: retunes each note toward its 5-limit JI ratio relative to Key via per-note detune (polyphonic), Amount blending from equal temperament to pure - the beatless thirds and fifths of real harmony.

└ **Microtuner** [TUN] [MIDI] in: Audio out: MIDI

Stretched-octave microtuning via per-note pitch detune.

└ **TuneTable** [TUN] [MIDI] in: Audio out: MIDI

Custom microtuning: detunes each note by a per-pitch-class cents offset from the node's 12-value config list (relative to Key) - meantone, Pythagorean, maqam or gamelan, any temperament. Amount scales the table.

Voicing (5)

⊗ **MidiUnison** [VOI] [MIDI] in: Audio out: MIDI

Doubles each note into multiple voices, optionally spread across channels.

⊗ **Mono** [VOI] [MIDI] in: Audio out: MIDI

Monophonic note priority (last / lowest / highest).

⊗ **MPEConvert** [VOI] [MIDI] in: Audio out: MIDI

Assigns each note its own rotating channel for per-note MPE expression.

⊗ **Slide** [VOI] [MIDI] in: Audio out: MIDI

Cthulhu's slide: mono legato glue. Picks one held note by Priority (last/low/high) and emits the new note-on before releasing the old one, so a portamento synth slurs between notes instead of retriggering.

⊗ **Sustain** [VOI] [MIDI] in: Audio out: MIDI

Sustain-pedal simulation: while Hold is engaged, note-offs are captured so the notes keep sounding; when Hold drops, every held note releases at once. Hold is a parameter, so a CC, an LFO or automation can play the pedal.

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ChordVoicing	6	MidiVibrato	5	TransArp	13
Counterpoint	6	Mono	14	Transformer	10
Diatonic	6	Mordent	12	Transpose	7
DiatonicChord	6	MPEConvert	15	Trill	14
Drone	6	NeoRiemann	7	TuneTable	14
Drunk	10	Note2CC	4	Tuplet	14
Echo	11	NoteFilter	9	Turn	14
EuclidMel	11	NoteLength	12	VelArp	14
Fall	5	NoteLimit	9	VelClip	4
FixedNote	9	NoteOffDelay	12	VelCompress	4
Flam	11	Octave	7	VelCurve	5
GateArp	11	OctaveArp	12	VelInvert	5
Generative	8	PatternGate	12	VelKey	5
Glissando	7	PitchDrift	8	VelLength	5
Harmonize	7	PolyChan	10	Velocity	5
Humanize	4	Polymeter	12	VelToCC	9
Invert	7	PressSwell	8	VoiceLead	7